The binary boundary for full-rank symmetry markings

Human Compression project: a continuation of the L031 correction

Let V be a vector space over $\mathbf{F}_2$ with quadratic form Q and polar form B. For $Q(x) = 1$ the symmetry in x is

$$s_x(v) = v + B(v, x)x.$$

It is an isometry, as substitution into the quadratic identity shows. It is nonidentity precisely when x is outside the polar radical. Write $N = \{x \in V : Q(x) = 1,\ x \notin \operatorname{rad} B\}$. We prove the binary nonexistence statement in all dimensions at least six. Together with the complete two-page companion *Three binary exceptions to a symmetry exclusion*, this determines exactly the existence boundary intended in Theorem 1.4 of Brooksbank–Ferrara–Leemans, arXiv:1709.02219v2. The companion is required in full for the three positive examples.

Lemma 1. *Suppose* $\dim V = d \geq 6$ *and* $\dim \operatorname{rad} B \leq 1$. *Then* N *spans* V, *and the group generated by all* s_x, $x \in N$, *acts transitively on* N.

Proof. The rank r_B of an alternating form is even. Hence $r_B \geq 6$ under these assumptions. If $U \leq V$ has codimension c, then

$$\operatorname{rank}(B|_U) \geq r_B - 2c. \tag{1}$$

Indeed, extend a basis of U to one of V: adding c rows and c columns to its Gram matrix increases rank by at most $2c$.

We shall also use the following elementary observation. If $u, v \in U$ satisfy $B(u, v) = 1$, then for any $z \in V$,

$$Q(z) + Q(z + u) + Q(z + v) + Q(z + u + v) = 1. \tag{2}$$

Thus every affine coset of $\langle u, v\rangle$ contains a vector of quadratic value one. The brackets here denote linear span.

Suppose N were contained in a hyperplane U. By (1), $B|_U$ has positive rank; choose $u, v \in U$ with $B(u, v) = 1$. For $z \notin U$, partition $z + U$ into its 2^{d-3} affine cosets of $\langle u, v\rangle$. Equation (2) gives at least $2^{d-3} \geq 8$ distinct vectors of quadratic value one outside U. At most one of these can be in the polar radical, since that radical has dimension at most one. At least one therefore belongs to N, a contradiction. Every proper subspace lies in a hyperplane, so N spans V.

Let $x, y \in N$ be distinct. If $B(x, y) = 1$, then $Q(x + y) = 1$ and $s_{x+y}(x) = y$. Also $x + y$ is outside the radical, as its polar product with x is one. If $B(x, y) = 0$, consider the simultaneous equations

$$B(x, z) = B(y, z) = 1. \tag{3}$$

The two linear functionals are nonzero. Over $\mathbf{F}_2$ they are either independent or equal. Hence (3) is consistent in both cases, and its solution set is an affine translate of a subspace U of codimension at most two. By (1), $B|_U$ has rank at least two. Choose $u, v \in U$ with $B(u, v) = 1$. Applying (2) in any four-point affine coset inside (3) supplies a solution z with $Q(z) = 1$. It is outside the radical since $B(x, z) = 1$. The first case now gives symmetries taking x to z and z to y. Their composition proves transitivity. □

Theorem 2. *Let* $d \geq 6$ *and* $\dim \operatorname{rad} B \leq 1$. *The full isometry group of* (V, Q) *cannot be generated by* d *nonidentity symmetries* $s_1, \ldots, s_d$ *satisfying* $[s_i, s_j] = 1$ *for* $|i - j| > 1$. *In particular it has no rank-*d *string C-group representation by symmetries.*

Proof. Suppose such generators exist, and choose $x_i \in N$ with $s_i = s_{x_i}$. Put $W = \langle x_1, \ldots, x_d \rangle$. Each generator preserves W and induces the identity on V/W, by the displayed symmetry formula. Every element of the generated group therefore induces the identity on V/W.

For every $x \in N$, the full isometry group contains s_x. Since x is nonradical, some v satisfies $B(v, x) = 1$, and $(s_x - 1)v = x$. Trivial action on V/W thus gives $x \in W$. Lemma 1 implies $W = V$. The d vectors x_i are consequently a basis.

For distinct basis vectors x_i, x_j of quadratic value one, $B(x_i, x_j) = 1$ prevents their symmetries from commuting: $s_i s_j(x_i) = x_j$, whereas $s_j s_i(x_i) = x_i + x_j$. Nonadjacent commutation therefore makes all Gram entries outside the two adjacent diagonals zero. Form the graph on the basis indices with an edge ij when $B(x_i, x_j) = 1$. It is a subgraph of the path $1, \ldots, d$.

If that graph were disconnected, the spans of its components would be a direct sum of nonzero orthogonal subspaces. Every generator preserves each such subspace, fixing those not containing its root. Their generated group could not take one basis vector to a basis vector in another component. But it is the full isometry group, which acts transitively on N by Lemma 1. This is a contradiction. The graph is connected, so

$$B(x_i, x_j) = 1 \quad \Longleftrightarrow \quad |i - j| = 1.$$

Put $n = d + 1$. In the even-weight space

$$E_n = \{z \in \mathbf{F}_2^n : \textstyle\sum_j z_j = 0\}, \qquad Q_n(z) = |\operatorname{supp} z|/2 \pmod 2,$$

the vectors $v_i = e_i + e_{i+1}$ form a basis with precisely these quadratic values and Gram entries. A quadratic form over $\mathbf{F}_2$ is determined by its values on a basis and these polar products: expand $Q(\sum t_i x_i)$ using $\sum_i t_i Q(x_i) + \sum_{i<j} t_i t_j B(x_i, x_j)$. Thus $x_i \mapsto v_i$ is an isometry $V \longrightarrow E_n$. In this model s_i is the adjacent coordinate transposition $(i, i+1)$. Hence the generated group consists exactly of the coordinate permutations S_n.

Since $n \geq 7$, choose a vector u with support of size six, a coordinate a in its support and a coordinate b outside it. Then $Q_n(u) = 1$, while $z = e_a + e_b$ satisfies $B(z, u) = 1$. The isometry s_u sends this weight-two vector to $z + u$, of weight six. Every coordinate permutation preserves weight. Therefore s_u is outside the proposed full group S_n, the final contradiction. □

The corrected binary existence statement. The companion proves full-isometry symmetry markings of ranks three, four and five, respectively on E_4, E_5, E_6, with the full intersection property. Their polar radicals have dimension one, zero and one, exactly the conditions of the supplied theorem. Theorem 2 proves that no other dimension $d \geq 3$ admits such a full-rank marking over $\mathbf{F}_2$. Thus the existential binary exception set is exactly $\{3, 4, 5\}$. This does not classify all individual quadratic forms or all their markings in those three dimensions.

The source's path argument is retained and made applicable to every possible full-rank generating set by the spanning and transitivity lemma. The weight-six symmetry supplies the explicit obstruction in higher dimension. These are elementary written arguments; no finite program, external group classification or Lean declaration is a premise. The positive larger-field theorem and its rank corollaries remain separate obligations in the full L031 recovery. There is no whole-original, open-conjecture or priority credit from this note.